

Lab 4: Normalization

CS355/CE373 Database Systems

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1 Instructions

- This lab will contribute $\approx 1\%$ towards the final grade.
- **The deadline for this lab is the end of your lab.**
- The lab must be submitted online via CANVAS. You are required to submit a PDF file that contains the final ERD.
- The PDF file should be named as *Lab_04_aa01234.pdf* where *aa01234* will be replaced with your student id.
- **Files that don't follow the appropriate naming convention will not be graded.**

1.1 Marking scheme

This lab will be marked out of 100.

- 50 Marks: completion of the lab.
- 50 marks: Progress and attendance (validated via viva).

1.2 Late submission policy

You can submit late till 11:59 PM on the same day as your lab with a 20% penalty. No submissions will be accepted afterward.

1.3 Use of AI

Taking help from any AI-based tools such as ChatGPT is strictly prohibited and will be considered plagiarism. Course staff may call students for Viva in case they feel that **AI** has been used for doing the lab.

2 Objective

The objective of this lab is to enable students to build data models for moderately complex business specifications, as per the rules of data normalization. The students should be able to assess and make sure that their data model does not leave room for anomalies in the dataset.

3 Exercise

Scenario 1 - Student Extracurricular Activities

Imagine you are employed as a Data Analyst at Habib University responsible for managing students' extracurricular activities information. A student can be a part of multiple extracurricular activities. Each of these activities are scheduled at different timings. During each time slot, there is always an Activity Lead who overlooks all the participating students in that group. A student can attend any of the time slots, as per their availability, so no specific slot will always have the same set of students. The university administration is using a table to store this information. Some of the rows from that table are given in Table 1 for reference.

1. Please analyze the provided extracurricular activities data table, and derive a relation schema for it in 1NF, by specifying the primary keys. Give reasons for choosing the primary keys.

- Identify full, partial, and transitive functional dependencies that exist among its attributes. If there are no dependencies, please state that explicitly.
- Normalize the schema up to the Third Normal Form (3NF) by creating new tables. Provide schemas for 2NF, and 3NF.

StudentID (PK)	StudentName	DOB	Year	Activity	ActivityType	ActivityTimings	ActivityLead	ActivityJoiningDate	ActivityLevel	GuardianID	GuardianName	GuardianContact
101	Ammara Javed	13-02-2005	1st	Basketball	Sports	Morning Afternoon	Mr. X Mr. Y	31-08-2024	Intermediate	1	Javed Jaffrey	555-1234
				Drama Club	Performance	Evening	Ms. A	15-09-2024	Beginner			
102	Bilal Syed	22-08-2004	2nd	Chess Club	Hobby	Morning	Mr. B	10-09-2023	Advanced	2	Mariam Syed	555-5678
				Coding	Educational	Afternoon	Mr. C	01-01-2024	Intermediate			
				Debate	Performance	Evening	Ms. D	15-06-2024	Beginner			
103	Dawood Waqar	10-09-2005	1st	Football	Sports	Morning Afternoon	Mr. E Mr. F	10-08-2024	Advanced	3	Sarah Bilal	555-4321
				Robotics	Educational	Morning	Mr. H	01-09-2024	Beginner			
						Evening	Mr. I					
104	Erum Pervez	20-01-2002	4th	Drama Club	Performance	Evening	Ms. A	01-09-2021	Advanced	4	Pervez Waheed	555-9876
				Cooking Club	Hobby	Afternoon	Ms. J	19-01-2024	Beginner			
105	Faheem Haq	23-05-2003	3rd	Debate	Performance	Afternoon	Ms. G	04-12-2023	Beginner	5	Nadeem Haq	555-6543
				Basketball	Sports	Morning	Mr. X	22-11-2022	Advanced			
						Afternoon	Mr. Y					
				Coding	Educational	Afternoon	Mr. C	07-10-2022	Advanced			
106	Ghazal Khan	24-07-2004	2nd	Chess Club	Hobby	Morning	Mr. B	26-10-2023	Intermediate	6	Mikaal Khan	555-3456
				Robotics	Educational	Morning	Mr. H	19-06-2024	Beginner			
						Evening	Mr. I					
107	Harish Chandra	01-01-2022	4th	Cooking Club	Hobby	Afternoon	Ms. J	29-04-2024	Beginner	7	Laxman Chandra	555-7890
				Football	Sports	Morning	Mr. E	18-03-2021	Advanced			
108	Ibrahim Shah	21-04-2003	3rd	Basketball	Sports	Afternoon	Mr. F			8	Adam Shah	555-6789
						Morning	Mr. X	12-12-2022	Intermediate			
109	Javeria Merchant	20-09-2004	2nd	Coding	Educational	Afternoon	Mr. C	17-11-2023	Intermediate	9	Ehtasham Merchant	555-1239
110	Kiran Zahid	29-03-2005	1st	Coding	Educational	Afternoon	Mr. C	30-08-2024	Beginner	10	Zahid Siddiqui	555-0123
				Drama Club	Performance	Evening	Ms. D					
111	Monis Shahmeer	16-04-2003	3rd			Evening	Ms. A	30-08-2024	Beginner	11	Shahmeer Zulfikar	555-2345
				Coding	Educational	Afternoon	Mr. C	15-06-2023	Intermediate			
						Evening	Ms. D					
112	Abdul Ahad	13-05-2002	4th	Football	Sports	Morning	Mr. E	18-12-2022	Advanced	12	Abdul Samad	555-4567
						Afternoon	Mr. F					
				Basketball	Sports	Morning	Mr. X	25-08-2021	Advanced			
						Afternoon	Mr. Y	27-08-2021	Advanced			
				Debate	Performance	Afternoon	Ms. G	01-01-2022	Intermediate			

Table 1: Student Extracurricular Activities

1NF is correct	15%
2NF is correct	30%
3NF is correct	35%
Functional and transitive dependencies are identified correctly.	20%

Table 2: Rubric for Scenario 1

You can also view the above table in Landscape mode at the end of this lab manual.

Scenario 2 - Atrium Cinema

You have to build an ERD for the given scenarios in DB Designer. The ERDs must be normalized up to 3NF and must contain the following details:

- Entities
- Attributes and their data types, null/not null constraints
- Primary Keys, Foreign Keys
- Relationships (1-1, 1-M)

You are building a database application for 'Atrium Cinema'. You could extract the following details from their website and have to design a data model that accommodates this information.

- There are multiple cinemas (A, B, and D) at Atrium with different screen sizes and capacity. A snapshot of the schedule of movies at each cinema is shown below:

SHOWTIMES FOR FRIDAY 13 MAY, 2016								
CINEMA A			CINEMA B			CINEMA D		
Show Time	Format	Movie	Show Time	Format	Movie	Show Time	Format	Movie
05:15 PM	2D	Hotel	03:00 PM	2D	1920 london	02:00 PM	3D	Captain America: Civil War
08:00 PM	2D	Mah-e-Mir	05:30 PM	2D	1920 london	05:00 PM	3D	The Angry Birds Movie
11:00 PM	2D	Baaghi	08:00 PM	3D	Captain America: Civil War	07:15 PM	3D	The Jungle Book
			11:00 PM	2D	1920 london	09:30 PM	3D	The Angry Birds Movie
						11:40 PM	3D	Captain America: Civil War

SHOWTIMES FOR SATURDAY 14 MAY, 2016								
CINEMA A			CINEMA B			CINEMA D		
Show Time	Format	Movie	Show Time	Format	Movie	Show Time	Format	Movie
11:30 AM	2D	Baaghi	11:45 AM	2D	1920 london	02:00 PM	3D	The Angry Birds Movie
05:15 PM	2D	Hotel	02:30 PM	3D	Captain America: Civil War	04:15 PM	3D	Captain America: Civil War
08:00 PM	2D	Mah-e-Mir	05:30 PM	2D	1920 london	07:15 PM	3D	The Angry Birds Movie
11:00 PM	2D	Baaghi	08:00 PM	3D	Captain America: Civil War	09:30 PM	3D	The Jungle Book
			11:00 PM	2D	1920 london	11:40 PM	3D	Captain America: Civil War

Figure 1: Cinema Shows

- Clicking on the name of a movie takes you to a Details page that gives the following information:

Country	Finland, USA
Year	2016-05-20
Category	Comedy, Action, Family, Animation
Release Date	May 20, 2016
Director	Clay Kaytis, Fergal Reilly
Cast	Jason Sudeikis, Josh Gad, Danny McBride, Maya Rudolph, Kate McKinnon, Sean Penn, Tony Hale, Keegan Michael Key, Bill Hader, Peter Dinklage
Producer	John Cohen, Catherine Winder

Figure 2: Movie Details

- You can book your ticket(s) online by providing the following information, and after booking, a seat number will be allocated to each ticket

Which Movie Do You Want To See?

City	Movie	Cinema	Date	Time	 Book
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Figure 3: Ticket Booking

Book a ticket and have fun movie time

Rs.. 500 / Seat	No. of Seats: 2	Total: 1000
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Seats: 2

Please Enter number of tickets you want to buy

2

Seat Preference:

Back

▼

Front
 Middle
 Back

Continue

Figure 4: Seat Selection

- The ticket prices may vary for different halls and the type of movie as shown below:

Ticket Price

Cinema A and B

2D Movies - Rs. 400 per person

3D Movies - Rs. 500 per person

Cinema D

2D Movies - Rs. 500 per person

3D Movies - Rs. 600 per person

Figure 5: Ticket Price

- Besides the current schedule of movies, the website also shows the list of movies to be shown in near future



Figure 6: Future Shows

Entities correctly identified	20%
All Attributes mentioned	15%
Relationships drawn and resolved correctly	20%
Correct Cardinalities correctly	10%
PK, FK identified and FK is placed in appropriate Entity table	10%
ERD is 3NF	25%

Table 3: Rubric for Scenario 2